ESM

# PLANT AUDIT REPORT

REF: AUD-000001 | COMPREHENSIVE PLANT AUDIT

DRAFT REPORT

**COMPANY DETAILS**

Name: Neptune Plast Machinery Ltd.
Code: NEPTUNE
Industry: Plastics & Industrial Machinery
Address: Plot 45-48, GIDC Industrial Estate, Makarpura, Vadodara, Gujarat

**PLANT / FACILITY**

Name: Makarpura Vadodara Manufacturing Unit
Location: Vadodara, Gujarat
Plant Code: PLANT-VAD-01
Contact: Sunil Mehta (Plant Head)

**AUDIT METRICS**

Auditor: Priya Narayanan
Status: In Progress
Completion: 65%
Audit Date: 13/7/2017

## 1. Executive Summary & Section Summary

| SEC | AUDIT SECTION | TOTAL | RECORDED | STATUS |
|---|---|---|---|---|
| A | A. Equipment & Facilities | 38 | 29 | In Progress |
| B | B. Plant Management / Leadership | 3 | 2 | In Progress |
| C | C. Plant Labor | 15 | 12 | In Progress |
| D | D. Capacity Utilization | 22 | 13 | In Progress |
| E | E. Plant Maintenance | 7 | 5 | In Progress |
| F | F. Quality | 7 | 6 | In Progress |
| G | G. Environmental Health and Safety | 9 | 7 | In Progress |
| H | H. Continuous Improvement | 7 | 7 | Complete |
| I | I. Regulation Fulfillment and Execution of Studies | 8 | 7 | In Progress |
| J | J. Energy, Electrical & Utilities (Suggested Additions) | 14 | 0 | Pending |
| K | K. Digital Systems & Audit Follow-up (Suggested Additions) | 6 | 0 | Pending |

## 3. Detailed Checkpoint Findings & Recorded Notes

**[Q006] Section A - Assessment: Compliant**

Notes / Findings: This factory was started in 2003 and currently has 16 operating lines making 12 & 16 mm emitting pipes and lateral pipes . Also has assembly for drippers/ventury, the drippers are imported from Israel and there is no moulding facility . The factory in Chennai has 5 lines only manufacturing lateral pipes with no assembly facility for emitting pipes. the Vadodara facility currently manufactures around 862 million meter of emitting/lateral pipes . They do not manufacture HDPE pipes required for sprikler , both HDPE/PVC pipes are outsorced . The sanctioned load is 2000 KVA with utilization of 1900 KVA, there is no UPS/DG back up since the power quality is very stable with dedicated power supply from the Electricity boards with no failures. Total RM consumption is 6000 MT ( LDPE/LLDPE) out of which 70% is purchased locally from IOCL/Reliance. The procurement spend $ 72 M ( 30% for dripper, 40% for RM and 15 % others like PVC/HDPE pipes). The transportaion cost is 2.7% of the sales

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q007] Section A - Assessment: Compliant**

Notes / Findings: RM Storage - 6 number silos of 70 MT each = 420 MT of raw material can be stored in silos , the feeding to silos is done thru the debagging station. RM conveying to each line is done auto feeders, Volmetric mixers are installed on each line for mixing of various componenents, each line is equipped with weight per meter control device like gravimetric

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q008] Section A - Assessment: Compliant**

Notes / Findings: 16 lines, unit 1 - 10 lines and Unit 2 - 6 lines . Unit 1 was started with 3 lines in 2003 and subsequently expanded to 10 lines . Unit 2 was set up in 2012 with 6 line operation

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q009] Section A - Assessment: Compliant**

Notes / Findings: Most of the extruders are KABRA INDIA make with outputs of 200 to 350 kg/hr . The downstream equipment's including the auto coilers are supplied from Netafim , Israel. Average line speed is 70 Meters/min. The age of machine is as follows - 3 lines are 15 years, 7 lines between 10 to 15 years and 6 lines are 5 years.

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q011] Section A - Assessment: Compliant**

Notes / Findings: Well maintained and upgraded continously , break down in under 1%

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q014] Section A - Assessment: Compliant**

Notes / Findings: Premises is on lease of 99 years

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q017] Section A - Assessment: Compliant**

Notes / Findings: Warehouse is around 30,000 square feet for accessories/bought out items/RM etc and is within the plant site

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q018] Section A - Assessment: Compliant**

Notes / Findings: FG storage is in open area

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q020] Section A - Assessment: Compliant**

Notes / Findings: Capital investment of $ 320 K in 2017 for upgrading the downstream machines and controls, to make line flexible for size changeovers etc. With this investment the productivity levels are expected to increase by 10-15%

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q022] Section A - Assessment: Compliant**

Notes / Findings: Pre fabricated building

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q023] Section A - Assessment: Compliant**

Notes / Findings: Metal roofing

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q024] Section A - Assessment: Compliant**

Notes / Findings: Yes

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q025] Section A - Assessment: Compliant**

Notes / Findings: Yes

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q026] Section A - Assessment: Compliant**

Notes / Findings: 7 numbers

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q027] Section A - Assessment: Compliant**

Notes / Findings: Sabic Research centre, Thermax, Du pont, Penwalt

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q029] Section A - Assessment: Compliant**

Notes / Findings: Brick walls upto 5 meter , after that prefabricated

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q030] Section A - Assessment: Compliant**

Notes / Findings: metal roofing

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q031] Section A - Assessment: Compliant**

Notes / Findings: Tremixing VDF Floors

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q032] Section A - Assessment: Compliant**

Notes / Findings: Conference room for 10

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q033] Section A - Assessment: Compliant**

Notes / Findings: Adequate

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q034] Section A - Assessment: Compliant**

Notes / Findings: Canteen facility for all employees

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q036] Section A - Assessment: Compliant**

Notes / Findings: water requirement is 1500 lits per day and is met  by city network, no borewell

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q037] Section A - Assessment: Compliant**

Notes / Findings: Own STP plant and the treated water is used for gardening

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q040] Section A - Assessment: Compliant**

Notes / Findings: 33174 Square meter

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q041] Section A - Assessment: Compliant**

Notes / Findings: 60% is built up area

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q043] Section A - Assessment: Compliant**

Notes / Findings: 5000 Square meter

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q044] Section A - Assessment: Compliant**

Notes / Findings: 3000 Square meter

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q046] Section A - Assessment: Compliant**

Notes / Findings: Vehicle parking is available for employees vehicles

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q047] Section A - Assessment: Compliant**

Notes / Findings: Unit 2 has space to accommodate another 4 extrusion lines

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q049] Section B - Assessment: Compliant**

Notes / Findings: The Plant Manager and SCM Head are well experinced with more than 20 years of relevant experience and are with company for more than 15 years.

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q050] Section B - Assessment: Compliant**

Notes / Findings: Average years in the company is 9-11 years

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q053] Section C - Assessment: Compliant**

Notes / Findings: Company employees in Manufacturing - 77 ( Production/Quality/Maint), Contract labours - 134 for operations like coiling/packing/shifting/grinding etc. Plant operates 315 days in a year . SCM has 25 employees

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q054] Section C - Assessment: Compliant**

Notes / Findings: NO Union

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q055] Section C - Assessment: Compliant**

Notes / Findings: N/A

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q056] Section C - Assessment: Compliant**

Notes / Findings: N/A

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q057] Section C - Assessment: Compliant**

Notes / Findings: N/A

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[N058] Section C - Assessment: Compliant**

Notes / Findings: N/A

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q059] Section C - Assessment: Compliant**

Notes / Findings: The contract workers are hired thru a contractor company , the wage negotiation is done with the contractor once in 3 years. Wages are higher than minimum wages and in line with area industry practices

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q060] Section C - Assessment: Compliant**

Notes / Findings: Average employee age is 28-31 years

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q061] Section C - Assessment: Compliant**

Notes / Findings: It is in line with the industry practice in the area

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q062] Section C - Assessment: Compliant**

Notes / Findings: Hardly any turnover

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q066] Section C - Assessment: Compliant**

Notes / Findings: Focus is on multiskilling training programme

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q067] Section C - Assessment: Compliant**

Notes / Findings: R & R ( REWARDS & RECOGNITION ) programme is in place for Kaizens and star performers

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q069] Section D - Assessment: Compliant**

Notes / Findings: Apparantly there does not seem any major bottleneck, the plant is working on Lean manufacturing practises in each area and showing improvements by value stream mapping of each area

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q070] Section D - Assessment: Compliant**

Notes / Findings: The key metrics which are mapped are like OEE, scrap %, meters production, units/meter etc . These are measured and monitored on real time basis with the help of automation system ( MII) and are benchmarked with other global facilties . The current ( OEE/OEU) of the factory is 71% and they have target and action plan for improving to 80%. Al the KPI'S are well displayed on the notice boards in the plant, the current scrap levels are under 5% and they have plans to reduce it further

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q074] Section D - Assessment: Compliant**

Notes / Findings: 10-15%, ( 862 mm to 988 mm)

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q077] Section D - Assessment: Compliant**

Notes / Findings: Energy units is 22 KWH/meter

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q079] Section D - Assessment: Compliant**

Notes / Findings: Calender hours less no plan

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q080] Section D - Assessment: Compliant**

Notes / Findings: Downstream equipments like coilers

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q081] Section D - Assessment: Compliant**

Notes / Findings: 70% of the volume is in 16 mm pipes, hence it helps in continous running of 7-8 machines with little size changeovers and reduced scrap levels. For balance 30% volume forecasting and planning plays important role in reducing the size changovers

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q083] Section D - Assessment: Compliant**

Notes / Findings: Yes preventive maintenance schedules are made and implemented for all lines. The braekdown is under 1%

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q085] Section D - Assessment: Compliant**

Notes / Findings: Based on the maximum line speed of each line in a particular size, means the line speed of 12 mm will be higher than 16 mm pipes. The average line speed is 70 MPM

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q086] Section D - Assessment: Compliant**

Notes / Findings: Yes there is lean team working on the various productivity improvement projects including value engineering

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q088] Section D - Assessment: Compliant**

Notes / Findings: As per IS quality plan

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q089] Section D - Assessment: Compliant**

Notes / Findings: The scrap/OEE/weight per meter etc are tracked real time thru automation on the machines ( thru PLC) and displayed real time

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q090] Section D - Assessment: Compliant**

Notes / Findings: As above

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q092] Section E - Assessment: Compliant**

Notes / Findings: Plant has well established preventive maintenance plan

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q093] Section E - Assessment: Compliant**

Notes / Findings: No

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q094] Section E - Assessment: Compliant**

Notes / Findings: The break down is under 1%

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q095] Section E - Assessment: Compliant**

Notes / Findings: Once in a month and during size changes

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q096] Section E - Assessment: Compliant**

Notes / Findings: Mostly inhouse

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q100] Section F - Assessment: Compliant**

Notes / Findings: They calculate cost of poor quality on monthly basis and it is showing downward trend on yearly basis. In addition they calculate line wise CP/CPK for improving quality

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q101] Section F - Assessment: Compliant**

Notes / Findings: All incoming material is approved as per test reports and the finished goods are checked /approved as per IS standards

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q102] Section F - Assessment: Compliant**

Notes / Findings: Yes they have QMS in place with test plans for incoming, in process and final inspection. They do not have third party inspection , the FG inspection is as per IS standards. Once in a year inspection by Govt of Gujarat for performance of drip system supplied

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q103] Section F - Assessment: Compliant**

Notes / Findings: Yes ISO 9001 - 2008 version, certified by International Certification Services with validity upto April 2018

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q104] Section F - Assessment: Compliant**

Notes / Findings: The lab is set up as per requirement of Indian standards and some QA equipment is as per Natafim Israel bencmarking including ESCR

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q105] Section F - Assessment: Compliant**

Notes / Findings: Yes

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q108] Section G - Assessment: Compliant**

Notes / Findings: They are tracking the TRIR/LTIR/SIR and other safety metrics. In 2016 and 2017 there is no recordable accident though there are cases of first aids

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q109] Section G - Assessment: Compliant**

Notes / Findings: NIL

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q110] Section G - Assessment: Compliant**

Notes / Findings: Focussed improvement plans for like safety audits, safety committee meetings, safety observation drives, daily shop floor meetings, electrical audits . All employees wear safety shoes , well laid yellow lines in place, machine guarding, no loose wiring, safety dashboards are in place, no water leakages

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q111] Section G - Assessment: Compliant**

Notes / Findings: They are certified with ISO 14001 but not with OHSAS 18001

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q112] Section G - Assessment: Compliant**

Notes / Findings: They  are certified with ISO 14001 - 2004 VERSION and validity is upto December 2018

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q113] Section G - Assessment: Compliant**

Notes / Findings: Yes they have safety committee which meets once in a month, they also have daily shop floor meetings where safety matters are discussed . There is designated place for having shop floor meetings

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q115] Section G - Assessment: Compliant**

Notes / Findings: The factory is situated in green zone belt with no pending voilation of Pollution and other regulations

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q118] Section H - Assessment: Compliant**

Notes / Findings: The key objectives and Indicators are 1) zero accidents 2) Cost of quality , reduction targets each year 3) Defined value engineering and productivity projects . Monthly operations review meeting and global bench marking. Lean journey started 2 years back LMA ( Lean maturity assessment ) has improved from 2.7 to current level of 4.3

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q119] Section H - Assessment: Compliant**

Notes / Findings: Yes the shop floor employees are involved in idea generation and improvements. The Kaizen programme is in place . Yes the members are suitably reawrded thru Kaizen programmes . They also have employee of the month and employee of the year programme to motivate the employees and internal talent search programme for taking higher responsibilities

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q120] Section H - Assessment: Compliant**

Notes / Findings: The shop floor workers are empowered

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q121] Section H - Assessment: Compliant**

Notes / Findings: N/A

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q122] Section H - Assessment: Compliant**

Notes / Findings: SAP R3 is implented along with Planning module and MII . The bar coding of products including that of ware house accessories etc in place. The add on modules are integrated with SAP

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q123] Section H - Assessment: Compliant**

Notes / Findings: Lean journey started in manufacturing and SCM in 2014. Lean deployment plan is made at the start of the year aong with deliverables, weekly meeting for review of progress , monthly steering committee meeting,

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q124] Section H - Assessment: Compliant**

Notes / Findings: The lean projects are being taken in manufacturing and SCM . Currently 15 lean projects are underway for improvement projects and each project is tracked for the implementaion and sustainence

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q125] Section I - Assessment: Compliant**

Notes / Findings: Yes - factory licence

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q126] Section I - Assessment: Compliant**

Notes / Findings: Yes - Pollution license

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q127] Section I - Assessment: Compliant**

Notes / Findings: Yes

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q129] Section I - Assessment: Compliant**

Notes / Findings: Yes

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q130] Section I - Assessment: Compliant**

Notes / Findings: Yes

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q131] Section I - Assessment: Compliant**

Notes / Findings: N/A

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

**[Q132] Section I - Assessment: Compliant**

Notes / Findings: N/A

*Auditor Remarks: Historical notes captured during site inspection on 13th July 2017.*

### 4. Reviewer Endorsement & Sign-Off

Auditor: Priya Narayanan (Emp ID: ESM-EMP-001)

Date of Audit: 13/7/2017

Submission Status: In Progress

Reviewer: Arun Kulkarni (Reviewer ID: ESM-REV-001)

Review Date: Pending Sign-Off

Decision: In Review / Draft